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A Physically Based Model for Soft Magnets' Anhysteretic Curve

JOSEFINA M. SILVEYRA [1,2,3] and JUAN M. CONDE GARRIDO [1,2]

1.—Laboratorio de Sólidos Amorfos, Facultad de Ingeniería, Universidad de Buenos Aires, Buenos Aires, Argentina. 2.—Instituto de Tecnologías y Ciencias de la Ingeniería "Hilario Fernández Long" (INTECIN), CONICET - Universidad de Buenos Aires, Buenos Aires, Argentina. 3.—e-mail: jsilveyra@fi.uba.ar

The push for more efficient electromagnetic devices for energy and functional applications requires improving the existing modeling approaches of their soft magnetic components. Following the parallel that exists between fluids and magnetism, we have developed a theory to explain the isothermal anhysteretic curve of soft magnetic materials. This curve depicts the ideal lossless demagnetization path that the volume magnetization would follow in the absence of changes in the size of the magnetic domains, by the action of the external field and the so-called molecular field. We offer a discussion on the physics behind the theory. A set of plots is proposed to reveal the magnetic homogeneity of a material and the number of component magnetizations in the model. We provide illustrative examples of a nanocrystalline alloy with different induced anisotropies and a grain-oriented electrical steel under different compressive stresses. An excellent agreement of our fittings with the experimental data taken from the literature demonstrates the scientific and technical usefulness of our approach for the magnetic materials community.

INTRODUCTION

Since the early years in which soft iron was used as a core for electromagnetic devices,[1] materials scientists have made important advances in the development of electrical steels,[2,3] soft ferrites,[4–6] soft magnetic composites,[7–9] and amorphous and nanocrystalline soft magnets.[10–13] These materials play a key role in developing highly efficient devices in the areas of power electronics[14,15] and electrical machines[16–18] due to the push for a more electrified and sustainable world.[19]

Thanks to later advances in computing capabilities, previous analytical models and physical prototypes of electromagnetic devices can now be complemented with numerical circuit simulations[20] and finite element analyses,[16,17] for which designers need simple and reliable descriptions of the magnetic behavior of soft magnets. Describing hysteresis in numerical models, however, is still very time-consuming for modern computers.[21] Conversely, modeling a constant magnetic susceptibility is simple, but it does not capture the saturation effect. Instead, the nonlinear anhysteretic curve is a good compromise between good accuracy and low computational cost. Two approaches have been used to describe the anhysteretic curve—with distinct physical interpretations. One approach defines it as the magnetization that would occur in the absence of material imperfections, while the other one defines it as the thermodynamic equilibrium state.[22] None of the direct measurement methods linked to these definitions seem practical to help in the design process of electromagnetic devices because they are tedious and time-consuming.[22–26]

Recently, we have reported studies where we computed the anhysteretic curve as the average curve of the left and right branches of the hysteresis loop[27,28] (i.e., for a given value of the magnetization M, we calculated the average value of the magnetic field H for both branches, as opposed to calculating, for a given value of H, the average value of M for both branches, which is the method described by Bozorth[29] to obtain an approximation of the "normal" or first magnetization curve). The challenge of using the average curve of the hysteresis loop is that numerical solvers need the magnetization data—and its first derivative—to be continuous

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and differentiable to avoid convergence problems. The typical strategy to achieve noise-free data is to use spline interpolation methods.[30] These are often combined with extrapolation methods to describe overfluxed regions in soft magnetic cores, facing then the challenge of achieving a reasonable approach to saturation.[30,31]

Instead of using interpolation and extrapolation methods for achieving a smooth anhysteretic curve, we recently reported a simple but accurate model that can easily describe the nonlinear magnetization over wide ranges of the applied field.[27,28] The utility of the method was demonstrated for isotropic and anisotropic, single- and two-component soft magnetic materials. The model extends the theory of Langevin and Weiss by allowing the molecular field to be either positive or negative, and by describing multicomponent systems by a law of partial magnetizations for magnetic mixtures that can be either aiding or opposing each other. Further studies remain to be carried out to clarify the physics behind the theory.

To fill this gap, we have used our method to fit and interpret the anhysteretic magnetization of two widely used soft magnets that present typically challenging modeling scenarios: an iron-based nanocrystalline alloy with different induced anisotropies, and a grain-oriented electrical steel under different compressive stresses.

This paper is organized as follows. In Section II, we describe our physically-based theory to explain the anhysteretic magnetization of soft magnets. We also offer a discussion about the component magnetizations, the molecular field constant, and the definition of the anhysteretic curve. Next, we present a set of plots that help unravel the magnetic homogeneity of a material, together with example curves that depict different scenarios. Section IV provides a step-by-step method to find the model parameters that fit a data curve. We then report the anhysteretic fittings optimized for the nanocrystalline and steel materials, and discuss the retrieved model parameters. The conclusions are set out in the last section.

THE EQUATION OF STATE FOR A MIXTURE OF MAGNETIC ENTITIES

Following the parallelism that exists between fluids and magnetism, Weiss found inspiration in van der Waals' theory of real gases[32] to extend Langevin's theory of paramagnetism,[33] and to describe the salient facts of ferromagnets.[34] Van der Waals had considered in his equation an internal pressure to account for the mutual interaction of molecules, so Weiss suggested an internal magnetic field to account for the mutual interaction among the elementary magnetic entities. With the same line of thought, we found inspiration in Amagat's and Leduc's law of gas mixtures[35–37] to extend Weiss's theory to describe the behavior of nonhomogenous magnets. Amagat and Leduc had proposed a law of partial volumes (which was more accurate for real gases than Dalton's earlier theory of partial pressures[38,39]), so we formulated a law of partial magnetizations.[28] Within this law of mixtures, the magnetization of the mixture (M) is the sum of the magnetization of its components (M_i), each at the applied magnetic field (H) and temperature (T) of the mixture:

$$M(H,T) = \sum_i M_i(H,T). \tag{1}$$

These component magnetizations are described in terms of Weiss' theory, which treats the medium as a set of magnetic entities uniformly distributed within a mesoscopic unit volume:

$$M_i = M_{Si} m_i, \tag{2}$$

$$m_i = \mathcal{L}(h_i) = \coth(h_i) - h_i^{-1}, \tag{3}$$

$$h_i = \frac{H + \alpha_i M_{Si} m_i}{a_i}. \tag{4}$$

In the equations above, m_i and h_i are, respectively, the component reduced magnetization and the reduced magnetic field. $\mathcal{L}$ is the so-called Langevin function, which assumes that the magnetic entities are regular pseudo-domains with the same fixed effective magnetic moment and size. For each component, we have three fitting parameters: the saturation magnetization (M_{Si}), the molecular field constant (α_i), and the parameter related to the pseudo-domain density

$$a_i = \frac{N_i k_B T}{\mu_0 M_{Si}}, \tag{5}$$

with N_i being the number of magnetitic entities per unit volume (or pseudo-domain density), k_B the Boltzmann constant, and μ_0 the magnetic vacuum permeability.

Therefore, the equation of state that describes the behavior of magnetically homogeneous materials is the single Langevin–Weiss function:

$$M = M_S m = M_S \mathcal{L}\left(\frac{H + \alpha M_S m}{a}\right), \tag{6}$$

while double-component magnetic mixtures are described by the double Langevin–Weiss function equation of state:

$$M = M_{S1} m_1 + M_{S2} m_2 = M_{S1} \mathcal{L}\left(\frac{H + \alpha_1 M_{S1} m_1}{a_1}\right) + M_{S2} \mathcal{L}\left(\frac{H + \alpha_2 M_{S2} m_2}{a_2}\right). \tag{7}$$

The component magnetizations may be aiding or opposing each other. Typically, the component magnetizations are aiding; however, we have recently predicted opposing magnetizations for certain cases.[28] We suggest that the theorized component magnetizations opposing the applied field are related to "local demagnetizing fields," earlier

proposed by Perevertov et al. to explain complex magnetic behaviors with maximum susceptibilities at nonzero applied fields.[40]

Historically, the molecular field constant had always been positive. In the analogous fluid theory, van der Waals had considered that, in equilibrium, the pressure of the gas would be equal to the external pressure, plus an internal pressure caused by attractive interactions between molecules. Then, Weiss reasoned that the magnetic field acting on the magnetic entities of a ferromagnet should be the sum of the applied field plus an internal field caused by the interactions between them. He named the internal field the "molecular field" and proposed it to be directly proportional to M (eventually expanding it into higher-order odd terms[41]). However, three decades later, Hamaker argued that the van der Waals interaction "between two particles of the same material embedded in a fluid is always attractive, provided there is no marked orientation of the fluid molecules. If the particles are of different composition, the resultant force may be a repulsion."[42] Direct measurements of repulsive van der Waals forces have been reviewed, for example, by Tabor et al.[43] The existence of repulsion forces in the fluid theory led us to explore the concept of negative molecular field constants in the magnetic theory (that is, molecular fields that may oppose the magnetization and applied magnetic fields).

We have recently published a study where we allowed the molecular field constant to be either positive or negative, and found that negative values were needed to explain the behavior of a soft magnet with uniform anisotropy.[27] The dependence of the single Langevin–Weiss function on the molecular field constant can be illustrated by the dimensionless plot shown in Fig. 1.

Another argument that we believe has led to the positive α preconception is that Heisenberg had associated it, under some considerations, with the exchange integral, which should be positive for ferromagnetism to occur (and negative for antiferromagnetism) (pp. 131–133).[44–46] However, there is also a so-called "sign problem" of the exchange integral because, in real ferromagnets, orbital wave functions are far more complex than in the scenario analyzed by Heisenberg.[47] This means that the exchange integral argument does not mandate positive molecular field constants for real ferromagnets.

Rather than associating the molecular constant solely to exchange forces (pg. 130),[46] we link it to crystal electric forces that influence the direction of the effective magnetic moment of the magnetic entities or pseudo-domains (the plane of the electron orbitals orient relative to the crystal axis to minimize the crystal field energy, while the electron magnetic moments orient relative to the orbitals to minimize the spin–orbit energy[48]). These forces result in an internal field that may aid or oppose the applied magnetic field in magnetizing the

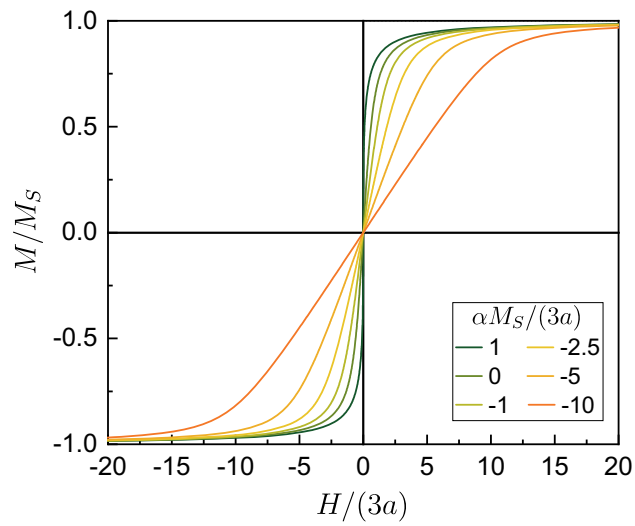

Fig. 1. Dimensionless plot of the single Langevin–Weiss equation of state (Eq. 6). The area to the left of the curve, related to the energy density stored in the material, increases as the molecular field constant (α) decreases.

material. Magnetic crystal forces opposing the action of the applied field have already been proposed in models describing the approach to saturation magnetization (pg. 486).[29] The demagnetizing factor, which accounts for dipolar interactions, is considered a constant contribution to α[49] (the effective demagnetizing factor depends on the external sample shape, which determines the macroscopic demagnetizing factor, and on the shape of single particles in, for example, soft magnetic composites, which determines the microscopic demagnetizing factor[50]). The cooperative phenomenon between magnetic moments–characteristic of ferromagnetism—would be implicit in the theory through the parameter a, which is related to the pseudo-domain density N. Indeed, compared to the high values found for paramagnets, ferromagnets present a very low a, which may be explained by magnetic moments coupling in parallel within large pseudo-domains. As a result, we may find ferromagnetic materials with negative molecular field constants and yet large magnetizations at low applied fields, as compared to paramagnetic materials (notice a is in the denominator of the reduced magnetic field (Eq. 4)). We should point out that, within this new definition of the molecular field constant, which can be negative, it cannot be related to the Curie temperature as in the Curie–Weiss theory (pg. 125).[46]

Our theory describes the isothermal anhysteretic magnetic curve differently from the two typical approaches mentioned in the Introduction.[22] We claim that the ideal lossless scenario is the demagnetization path that would take place, at a constant temperature, after achieving a certain magnetic state (H_{TIP}, M_{TIP}), by the action of the external field and the so-called molecular field, which is

proportional to the average magnetization of a mesoscopic unit volume. This ideal demagnetization path occurs in the absence of changes in the size of the pseudo-domains and, thus, in the size of the domains (see Fig. 2). The condition agrees with the observation of domain multiplication—instead of simple magnetization rotation—in hysteretic demagnetization processes.[51] In this work, we show the correlation between the pseudo-domain density of the model and the actual domain density present in a material.

THE SET OF PLOTS TO REVEAL A MATERIAL'S MAGNETIC HOMOGENEITY

We propose using a number of plots to readily reveal the magnetic homogeneity of a soft magnet and, therefore, the number of component magnetizations in the law of mixtures (Eq. 1). First, we plot the magnetization and the susceptibility ($\chi = \partial M/\partial H$) with respect to the applied field in the typical linear scales. Then, we represent the same data, but with a logarithmic scale for H. The beauty of semilogarithmic plots is that we can see details more easily for low as well as for high applied fields. Finally, the visual impression provided by the semi-log representation of the magnetization curve can be further enhanced by the introduction of the semi-log magnetization derivative ($\partial M/\partial(\ln H) = H\partial M/\partial H$), which represents the slope of the semi-logarithmic plot of M versus H (actually, directly proportional to such slope if we use a common logarithmic scale for H in the horizontal axis and compute the derivative of M with respect to the natural logarithm of H).

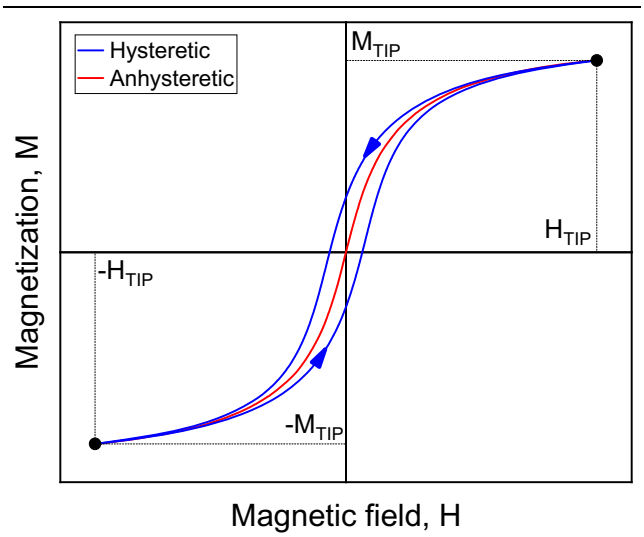

Fig. 2. Hysteretic demagnetization path a ferromagnetic material follows after achieving the magnetic state (H_{TIP}, M_{TIP}). If the material is subsequently magnetized up to ($-H_{TIP}, -M_{TIP}$), and back to (H_{TIP}, M_{TIP}), the paths describe the so-called hysteresis loop. The average curve of the left and right branches of the loop depicts the anhysteretic path proposed in this work.

Figure 3 shows the set of plots corresponding to the single Langevin–Weiss function for a given set of parameters (Eq. 6). This type of curve describes the class of materials with a homogenous magnetic nature at a mesoscopic scale such as, for example, soft ferrites and nanocrystalline alloys.[28] From these plots, we can readily identify one critical field (H_{cr}), at which the semi-log M derivative is maximum.

For a mixture of two aiding component magnetizations (Eq. 7), two critical fields in the semi-log M derivative plot reveal a material's magnetic inhomogeneity, as observed, for instance, for some steels.[28] Figure 4 illustrates an example scenario described by the double Langevin–Weiss function.

If the component magnetizations were to be, not adding, but opposing each other, the model predicts the type of curves shown in Fig. 5. In this case, it is the maximum susceptibility at a nonzero applied field that reveals the magnetic inhomogeneity of the material. This behavior has been observed for grain-oriented silicon steels measured either transversely to the rolling direction[52] or under compressive stresses.[53,54]

By finding the critical fields in the semi-log M derivative plots, together with the corresponding reduced magnetizations, and by imposing that the modeled curve passes through the tip point of the data curve (H_{TIP}, M_{TIP}), it is possible to obtain the model parameters that fit the data. This fitting method is summarized in the next section.

THE FITTING METHOD

The ability to retrieve the fitting parameters given a set of experimental data is a key step in the implementation of a magnetization model.[55] Typically, researchers have used stochastic methods to find the Langevin–Weiss function parameters, M_{Si}, α_i, and a_i, whose bounds are uncertain and generally need to be defined after several trials.[56,57] As an alternative to such time-consuming approaches that may even have convergence problems,[58] we have proposed transforming the search space to find H_{cri} and m_{cri} instead, which are intrinsically bounded.

In the following, we provide a practical recipe to fit anhysteretic magnetization curves of homogeneous and nonhomogenous materials (the demonstration can be found in Ref. 28).

Let us begin by describing the procedure for retrieving the three parameters of the single Langevin–Weiss function. First, the parameter a can be obtained from:

$$a = H_{cr}P(m_{cr})\left(Q(m_{cr}) \pm \sqrt{Q^2(m_{cr}) - 1}\right). \qquad (8)$$

This equation has two possible results that need to be examined (the smallest a describes the narrowest peak in the semi-log M derivative plot). The functions denoted as P and Q are given, respectively, by:

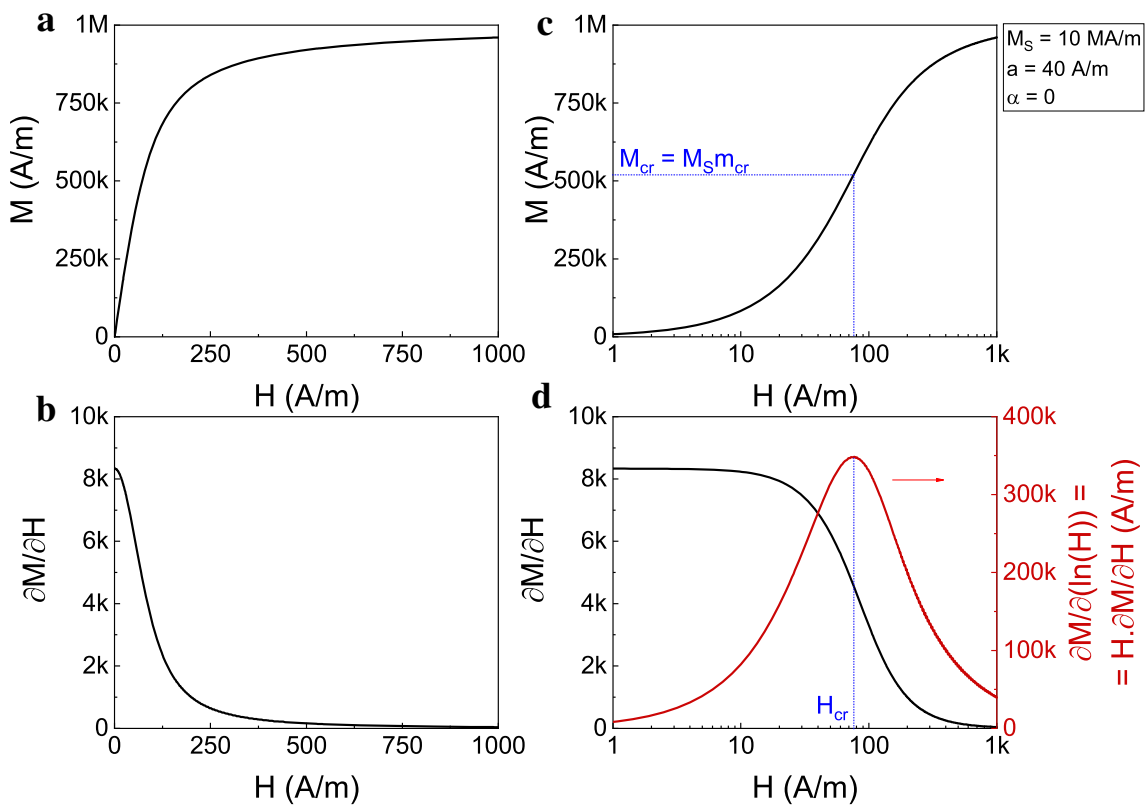

Fig. 3. Example curves corresponding to a magnetically homogeneous magnetic material, which follows the single Langevin–Weiss equation of state (Eq. 6). The proposed plots for the anhysteretic magnetization analysis are: the magnetization and the magnetic susceptibility with respect to the applied field with the typical linear representation (a and b, respectively), the same data but with a logarithmic scale for H (c and d), and the semi-log magnetization derivative plot (d, using the right vertical axis). The critical field (H_{cr}) at which the semi-log M derivative is maximum, together with corresponding magnetization (M_{cr}) are indicated in the plots.

$$P(m) = \frac{\mathcal{L}'\left(\mathcal{L}^{-1}(m)\right)}{m - \mathcal{L}^{-1}(m)\mathcal{L}'\left(\mathcal{L}^{-1}(m)\right)}, \tag{9}$$

$$Q(m) = \left(\frac{m}{\mathcal{L}'\left(\mathcal{L}^{-1}(m)\right)}\right)^2 \frac{\left(-\mathcal{L}''\left(\mathcal{L}^{-1}(m)\right)\right)}{2\left(m - \mathcal{L}^{-1}(m)\mathcal{L}'\left(\mathcal{L}^{-1}(m)\right)\right)} - 1, \tag{10}$$

where the Langevin inverse ($\mathcal{L}^{-1}$) may be evaluated using Kröger's approximant, as it offers simultaneous simplicity and accuracy (its maximum relative error is below 0.3%[59]):

$$\mathcal{L}^{-1}(m) \cong \frac{3m - \frac{m}{5}\left(6m^2 + m^4 - 2m^6\right)}{1 - m^2}. \tag{11}$$

The first and second derivatives of the Langevin function with respect to the reduced magnetic field are, respectively:

$$\mathcal{L}'(h) = \frac{\partial \mathcal{L}(h)}{\partial h} = -\mathrm{csch}^2(h) + 1/h^2 \underset{h\to 0}{\rightarrow} \frac{1}{3} - \frac{h^2}{15} + \frac{2}{189}h^4 - \ldots, \tag{12}$$

$$\mathcal{L}''(h) = \frac{\partial \mathcal{L}'(h)}{\partial h} = 2\coth(h)\mathrm{csch}^2(h) - 2/h^3 \underset{h\to 0}{\rightarrow} -\frac{2}{15}h + \frac{8}{189}h^3 - \frac{2}{225}h^5 + \ldots, \tag{13}$$

The Taylor expansions about the origin are preferred over the direct evaluation of the actual analytical expression in the vicinity of $h = 0$, because the functions suffer from catastrophic cancellation.

After obtaining a, the product αM_S can be calculated as:

$$\alpha M_S = \frac{\mathcal{L}^{-1}(m_{cr})a - H_{cr}}{m_{cr}}. \tag{14}$$

From Eqs. 8 and 14, it follows that the dimensionless product used in Fig. 1 can be written in terms of m_{cr} as:

$$\frac{\alpha M_S}{3a} = \frac{1}{3m_{cr}}\left(\mathcal{L}^{-1}(m_{cr}) - \frac{1}{P(m_{cr})\left(Q(m_{cr}) \pm \sqrt{Q^2(m_{cr}) - 1}\right)}\right) \tag{15}$$

The equation can be solved numerically for m_{cr} and is plotted in Fig. 6.

For a magnetically homogenous material, the saturation magnetization can be obtained by imposing that the modeled curve passes through the tip of the data curve (H_{TIP}, M_{TIP}):

$$M_S = M_{TIP}/m(H_{TIP}). \tag{16}$$

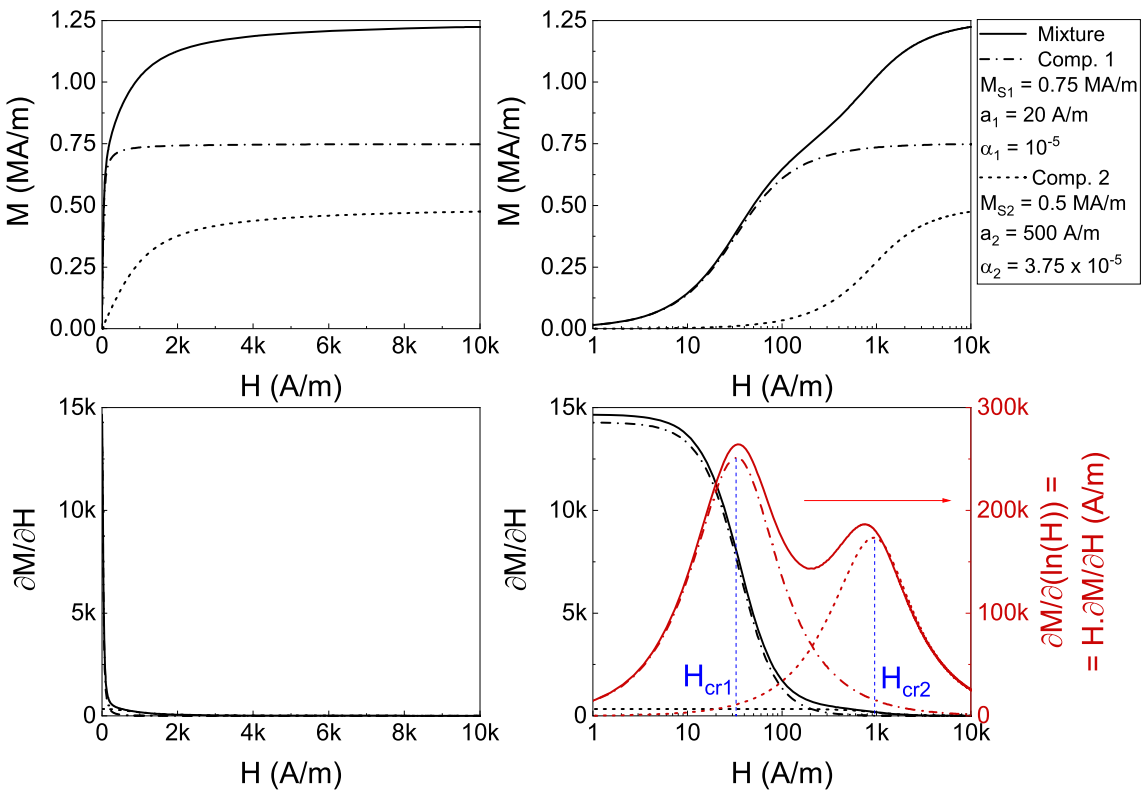

Fig. 4. Example curves corresponding to a magnetically nonhomogeneous magnetic material, which follows the double Langevin–Weiss equation of state (Eq. 7) with two aiding component magnetizations. Two critical fields (H_{cr1} and H_{cr2}) in the red semi-log magnetization derivative plot reveal the material's magnetic inhomogeneity (Color figure online).

The reduced magnetization at a given applied field (e.g., $m(H_{TIP})$ in the latter equation) can be calculated by finding the root of:

$$\mathcal{L}\left(\frac{H+\alpha M_S m}{a}\right) - m = 0, \tag{17}$$

over the starting range $0 < m < 1$ through the simple and robust bisection method (this method may be combined in a hybrid algorithm with the Newton–Raphson method for a more rapid convergence using the routine RTSAFE in Ref. 60).

Although not really necessary for computing the single Langevin–Weiss function, α can be calculated as:

$$\alpha = \alpha M_S / M_S. \tag{18}$$

Finally, we recommend performing the fitting by minimizing the orthogonal (also called diagonal) distance between the data curve and the modeled curve in a normalized M versus $\log(H)$ plane (see Appendix C in Ref. 28 for further details). At the expense of a higher computational cost, we obtained better fittings using this approach rather than with the conventional techniques of minimizing the root-mean-squared error in M at constant H or in H at constant M. To this end, a nonlinear solving method could be used (e.g., the generalized reduced gradient nonlinear solver). The two fitting parameters, H_{cr} and m_{cr}, are intrinsically bounded. H_{cr} must remain close to the center of the peak in the semi-log M derivative plot (Fig. 3). m_{cr} must be positive and below unity, and a reasonable initial seed value is around 0.5 (see Fig. 6). Since the a parameter must be a real number, by considering that the radicand in Eq. 8 must be nonnegative, we have found numerically that m_{cr} can be further constrained so that $m_{cr} \geq 0.4495$ (this minimum value can also be observed graphically in Fig. 6).

For a two-component mixture, we need to find two critical fields (H_{cr1} and H_{cr2}), together with their corresponding reduced magnetizations (m_{cr1} and m_{cr2}). For the opposing component magnetizations case, where only one initial seed for a critical field is revealed by the semi-log M derivative plot, experience has shown that a good initial seed for the second critical field is about 1/3 or 1/2 of the first one. The parameters a_1, a_2, $\alpha_1 M_{S1}$, and $\alpha_2 M_{S2}$ can be obtained by using Eqs. 8–14. Then, by imposing that the modeled curve passes through the tip and a second point of the data curve, M_{S1} and M_{S2} can be calculated as:

$$M_{S1} = \frac{m_2(H_X)M_{TIP} - m_2(H_{TIP})M_{data}(H_X)}{m_2(H_X)m_1(H_{TIP}) - m_2(H_{TIP})m_1(H_X)}, \tag{19}$$

$$M_{S2} = \frac{m_1(H_X)M_{TIP} - m_1(H_{TIP})M_{data}(H_X)}{m_1(H_X)m_2(H_{TIP}) - m_1(H_{TIP})m_2(H_X)}. \tag{20}$$

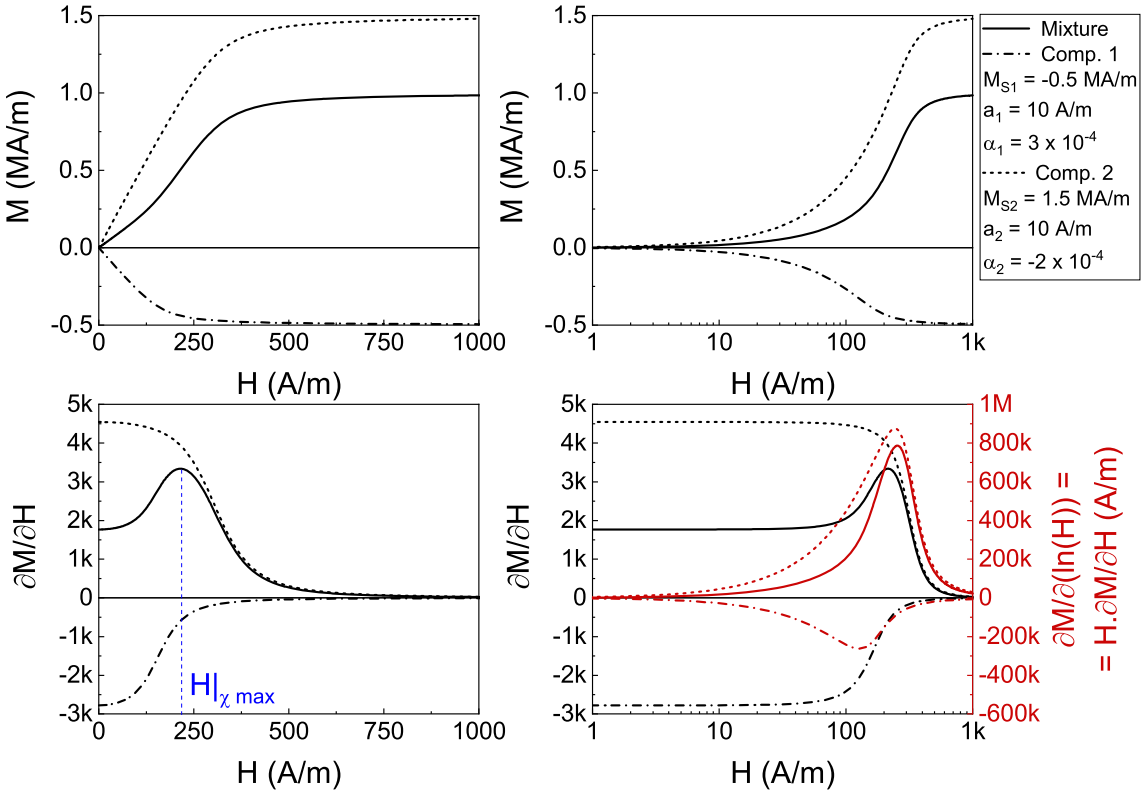

Fig. 5. Example curves corresponding to a magnetically nonhomogeneous magnetic material, which follows the double Langevin–Weiss equation of state (Eq. 7) with two opposing component magnetizations. The nonzero applied field at which the susceptibility is maximum ($H|_{\chi max}$) reveals the material's inhomogeneity.

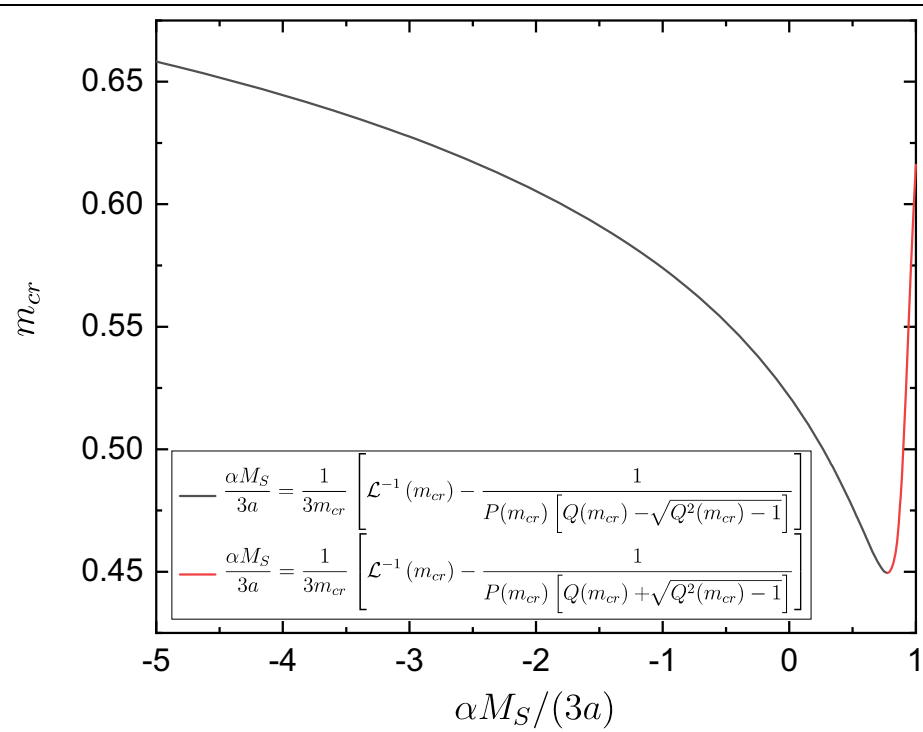

Fig. 6. Dependence of the reduced magnetization at the critical field on the dimensionless product $\alpha M_S/(3a)$. The legend indicates the inverse relation of each part of the curve.

H_X denotes the applied field at the second chosen point of the data curve, which may be conveniently selected between the two critical fields, but may also be optimized in the fitting. The corresponding magnetization, $M_{data}(H_X)$, can be obtained by interpolating the data. Finally, α_1 and α_2 can be solved from (Eq. 18).

In sum, instead of directly searching for the optimum model parameters (M_{Si}, a_i, α_i), which have uncertain bounds, we propose to search for other intrinsically bounded parameters (H_{cri}, m_{cri}). The approach presented here for double-component mixtures may be further extended for systems with even more components. However, deterministic optimization methods may be limited by local minima, which increase with the number of component magnetizations. Even though the retrieved parameters may serve as a phenomenological description of the material, a physical interpretation of the results requires a careful analysis, as for any multivariate fitting technique. A high number of component magnetizations will also increase the computation effort of $M(H)$. Yet, the utility of the method cannot be overemphasized: by just storing the corresponding model parameters in a materials library, the equation of state (Eq. 1) can be used to generate a look-up-table of M versus H to speed up runtime computations of a numerical solver. This precalculated magnetization data can have the desired precision and are continuous and differentiable, as also is its first derivative.

We should mention that, in a previous work,[27] we reported another possible fitting method. That approach is limited to magnetically homogeneous systems, but has the advantage of being faster and using a less cumbersome formulae. Rather than finding H_{cr} and m_{cr}, within this method, we need to find the reduced magnetization at the tip of the data curve (m_{TIP}, which must be positive and below unity) while imposing that the modeled curve passes through two points of the data curve (the tip and a second point). Then, the model parameters can be computed as:

$$M_S = m_{TIP} M_{TIP}, \tag{21}$$

$$\alpha = \frac{H_{TIP}}{M_{TIP}} \frac{\frac{\mathcal{L}^{-1}\left(\frac{M_X}{M_S}\right)}{\mathcal{L}^{-1}(m_{TIP})} - \frac{H_{data}(M_X)}{H_{TIP}}}{\frac{M_X}{M_{TIP}} - \frac{\mathcal{L}^{-1}\left(\frac{M_X}{M_S}\right)}{\mathcal{L}^{-1}(m_{TIP})}}, \tag{22}$$

$$a = \frac{H_{TIP} + \alpha M_{TIP}}{\mathcal{L}^{-1}(m_{TIP})} \tag{23}$$

Here, too, we need to evaluate $\mathcal{L}^{-1}$ using an approximant (Eq. 11). M_X denotes the magnetization at the second point of the data curve, which may be conveniently set to be about $0.7M_{TIP}$, but may also be optimized in the fitting. The corresponding applied field, $H_{data}(M_X)$, can be obtained by interpolating the data.

ANALYSIS AND INTERPRETATION EXAMPLES

We now illustrate our theory and fitting method by analyzing, for the first time, two soft magnetic materials with challenging modeling features, and yet widely used for energy applications and other inductive devices: an iron-based nanocrystalline alloy, under the tradename FINEMET® or VITROPERM®, with different magnetic field-induced anisotropies; and a grain-oriented silicon steel under different compressive stresses applied parallel to the rolling and measurement directions. Before the fitting analysis, we provide a brief description of both materials, which will help to understand the interest in the study and to interpret the obtained results.

Typical commercial grades of $Fe_{\sim 74}Si_{13\text{-}16}B_{6\text{-}9}Nb_3Cu_1$ as-quenched amorphous ribbons achieve a nanocrystalline state when subjected to a controlled annealing process (i.e., $\sim$ 20-nm-wide $Fe_{3-x}Si_x$ crystals magnetically coupled by a remaining amorphous matrix[61,62]). The simplest heat treatment is performed under no applied magnetic field and achieves a rounded hysteresis loop. Annealing the alloy in a magnetic field induces a uniaxial anisotropy that is uniform on a scale much larger than the exchange length. The uniaxial anisotropy has an easy axis along the annealing field direction that originates from a directional ordering of pairs of Si atoms within the grains along such an axis.[63–66] This induced anisotropy dominates over magnetoelastic and magnetocrystalline anisotropy contributions, so the resulting nanocrystalline state presents fewer property variations along the material (the alloy annealed without applied field is more sensitive to disturbing effects during nanocrystallization, such as stray fields and residual stresses).[64] The induced anisotropy also helps to tune the shape of the hysteresis loop to meet the demands of different applications.[64] Longitudinal and transverse field annealing yields hysteresis loops with large and small squareness ratios, respectively.[63] Soft magnets with rectangular loops present the highest initial susceptibilities and are suitable for saturable reactors,[67] ground fault interrupters, or common mode chokes.[64] In contrast, materials with flat loops can have very low susceptibilities that are almost constant over a wide field range. The magnetization process under a longitudinal field is governed by an almost reversible rotation of the magnetic moments within a domain, from the easy axis to the applied magnetic field axis.[64] Losses are correspondingly very low, even at high frequencies, which makes them particularly useful for high-frequency power transformers in switched power supplies and interface transformers in the ISDN-

telecommunication network.[64,67] In addition to the technological interest in these different magnetic behaviors, both longitudinal and transverse magnetic field annealing is still attracting researchers from a fundamental point of view.[68–70]

Grain-oriented electrical steels remain the materials of choice for power transformer cores. The manufacturing process of these devices inevitably causes some mechanical stress on the core that can strongly affect its magnetic properties and, therefore, the performance of the device.[71] With the increase of either compressive stresses parallel to the rolling direction or tensile stresses transverse to the rolling direction, hysteresis loops transform from a typical sigmoidal shape at zero stress to constricted loops with a two-peak differential susceptibility.[72,73]

For this study, we analyzed quasistatic hysteresis loops reported in the literature. The first set of curves corresponds to wound toroids of a $Fe_{73.5}Si_{13.5}B_9Nb_3Cu_1$ alloy annealed for 1 h at 540°C to induce its nanocrystallization: (1) without a magnetic field (i.e., no field annealing), (2) with a magnetic field applied parallel to the ribbon axis (i.e., longitudinal field annealing), and (3) first nanocrystallized at 540°C with no field and subsequently annealed at 350°C with a magnetic field applied transverse to the ribbon axis (i.e., transverse field annealing).[64,74] The hysteresis loops were measured along the circumferential direction of the toroids. The second set of curves corresponds to a grain-oriented silicon steel sheet, cut parallel to the rolling direction, longitudinally compressed up to 67 MPa.[71] The loops were measured along the rolling direction of the steel sheets, which corresponds to the surface-parallel easy axis. The plots of the experimental quasistatic hysteresis loops together with the fitted anhysteretic curves are shown in Figs. 7 and 8, and demonstrate the ability of our theory and method to capture the magnetic nonlinear characteristic in all these challenging scenarios.

To serve as an example of the performed analysis, we show in Fig. 9 the proposed set of plots with data and fitted curves for the nanocrystalline alloy with transverse field annealing. The anhysteretic data $M(H)$ was obtained by averaging the left and right branches of the hysteresis loop. The loop reported in the literature and shown in Fig. 7 had been presented as B versus H, so it had to be transformed into M versus H using $M = B/\mu_0 - H$. In Table I, we report the optimized critical field and corresponding reduced magnetization for the three types of annealing, together with the calculated model parameters. In the past, this magnetically homogeneous material with uniform induced anisotropy could not be modeled with the Langevin–Weiss function, because of the preconception of a mandatory positive molecular field constant.[56,75,76]

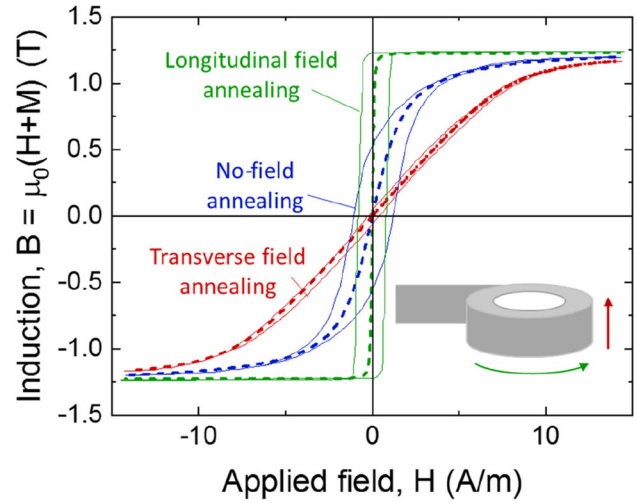

Fig. 7. Magnetic behavior of a wounded toroid of $Fe_{73.5}Si_{13.5}B_9Nb_3Cu_1$ alloy nanocrystallized under longitudinal field annealing, no-field annealing, and transverse field annealing. Solid curves correspond to the quasistatic hysteresis loops digitized from Ref. 64, while dashed curves correspond to the anhysteretic curves modeled with single Langevin–Weiss functions using our method.

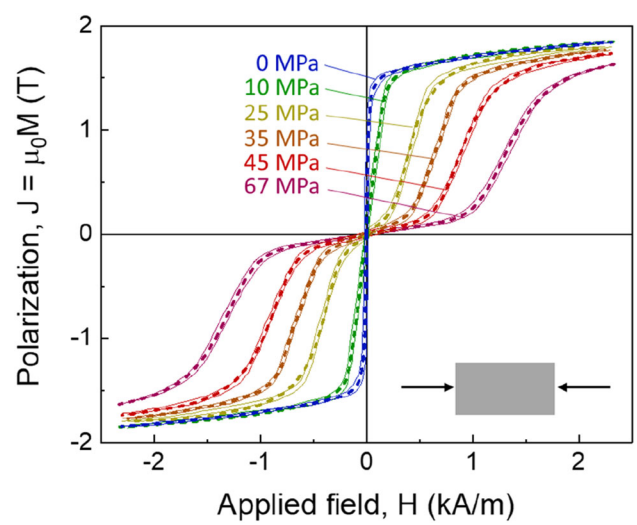

Fig. 8. Magnetic behavior of a grain-oriented silicon steel under compressive stresses. Solid curves correspond to the quasistatic hysteresis loops digitized from Ref. 71, while dashed curves correspond to the anhysteretic curves modeled with double Langevin–Weiss functions using our method.

Next, we show in Fig. 10 the proposed set of plots with data and fitted curves for the grain-oriented steel compressed at 67 MPa. The anhysteretic data $M(H)$ was obtained by averaging the left and right branches of the hysteresis loop. The loop reported in the literature and shown in Fig. 8 had been presented as J versus H, so it had to be transformed into M versus H using $M = J/\mu_0$. The susceptibility peak at a nonzero field reveals its nonhomogeneous nature. Using our fitting method, we were able to find the critical fields of two opposing magnetic components together with their corresponding

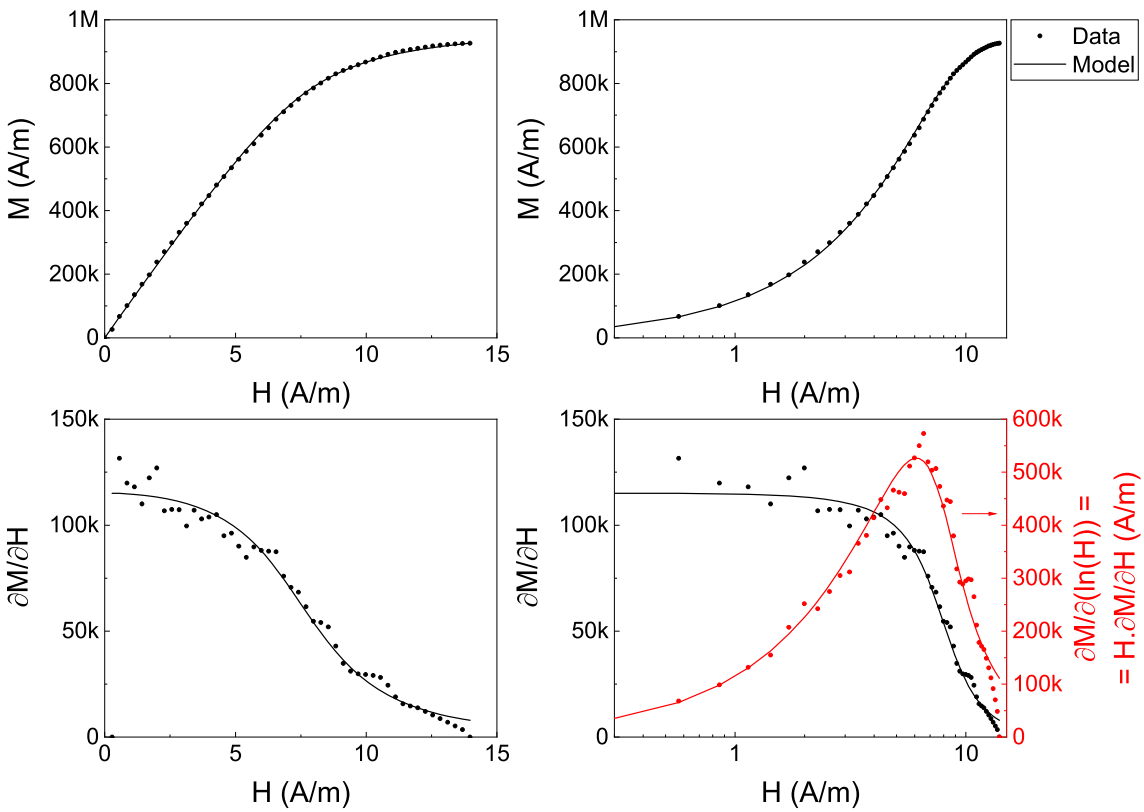

Fig. 9. Set of plots used to analyze the $Fe_{73.5}Si_{13.5}B_9Nb_3Cu_1$ alloy nanocrystallized under transverse field annealing and measured along the longitudinal direction of the ribbon.

Table I. Parameters of the anhysteretic magnetization curves parallel to the ribbon axis of a nanocrystalline Fe-based alloy under different magnetic field annealing conditions: the optimized critical applied field and corresponding reduced magnetization (H_{cr} and m_{cr}), the parameters of the single Langevin–Weiss function (M_S, a, α), the material saturation polarization (J_S), the dimensionless product used in Fig. 1 ($\alpha M_S/(3a)$), and the product related to the pseudo-domains density ($\mu_0 M_S a = Nk_BT$)

Parameter	Units	Longitudinal field	No field	Transverse field
H_{cr}	A/m	0.10	1.33	6.11
m_{cr}		0.63	0.51	0.66
M_S	kA/m	984	1,015	987
a	A/m	0.01	0.88	0.44
α		− 1.1E-07	5.4E-07	− 7.4E-06
$J_S = \mu_0 M_S$	T	1.24	1.27	1.24
$\alpha M_S/(3a)$		− 3.18	0.21	− 5.52
$\mu_0 M_S a$	J/m^3	0.01	1.12	0.54

reduced magnetizations, from which we calculated the 6 parameters of the double Langevin–Weiss function (see Table II).

For each of the fittings, we used the retrieved model parameters to compute the saturation polarization ($J_S = \mu_0 \sum_i M_{Si}$), the dimensionless product $\alpha_i |M_{Si}|/(3a_i)$, and the product $\mu_0 M_{Si} a_i$ (see Tables I and II).

First, we estimated J_S due to its relevance for technological applications. We estimated a saturation polarization of 1.24 T for field-annealed nanocrystalline alloys, in perfect agreement with the manufacturer's product information.[74] The alloy annealed without an applied field showed a slightly different value (1.27 T). Variations are expected in nanocrystalline alloys annealed with no field[11] and, as previously explained, are due to the material sensitivity to disturbing effects during the heat treatment, such as stray fields and residual stresses. For the grain-oriented electrical steel, we estimated a saturation polarization of ∼ 1.9 T. We are unable to determine from the available data[71] whether the differences in the estimated values respond to an effect of the compressive stress.

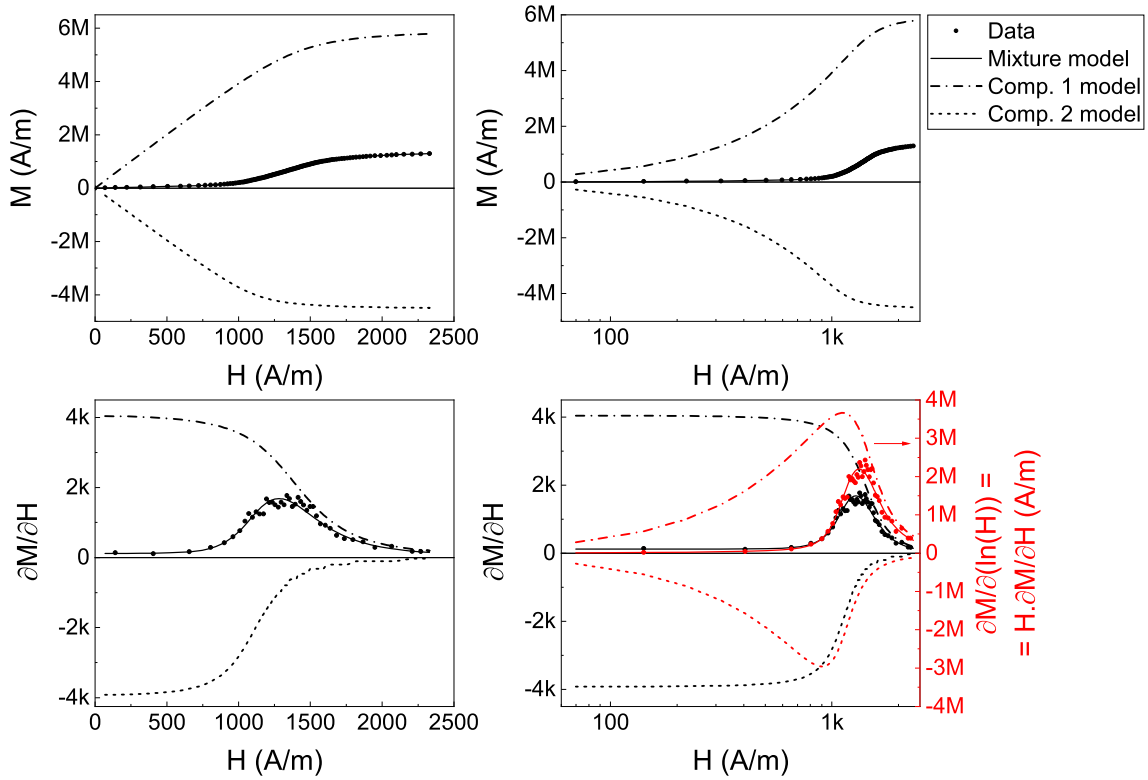

Fig. 10. Set of plots used to analyze the grain-oriented electrical steel under 67 MPa compressive stress measured along the rolling direction of the steel sheet.

Second, the reported dimensionless product is related to the anisotropy energy of the material and was used in the dimensionless plot of the Langevin–Weiss function (Fig. 1). The dimensionless product is, in absolute value, one order of magnitude larger for nanocrystalline alloys with induced uniform anisotropy than for the alloy without it (with opposite signs because $\alpha < 0$ for both field-annealed alloys). Highly compressed grain-oriented electrical steels (≥ 25 MPa) exhibit dimensionless products for both component magnetizations that increase almost linearly, in absolute value, with the increase of the applied stress. This suggests stronger crystal forces for stronger compressive stresses. Steels with lower stresses show a more erratic behavior with compression.

The third and last reported product is equal to $N_i k_B T$ (see Eq. 5) and, thus, is proportional to the number of magnetic entities of a component per unit volume. Let us recall that the elementary entities in the Langevin–Weiss theory are pseudo-domains within a magnetic component, all having equal magnetic moment and size. This means that pseudo-domains are not equivalent to the actual magnetic domains present in the material. However, we did find a qualitative correlation, for both types of materials, between the calculated pseudo-domain densities (see Tables I and II) and the reported surface domain observations at the demagnetized state, which we describe below.

The demagnetized domain structure of materials with uniform uniaxial anisotropy is particularly simple.[46,64] Nanocrystalline alloys annealed under a longitudinal magnetic field present regular wide slab domains parallel to the ribbon axis.[77] When annealed under a transverse magnetic field, the periodic slab domains are narrower and transverse to the ribbon axis.[78–81] Annealing without the application of a magnetic field does not mean that the alloy has no induced anisotropies: anisotropies are always induced along the direction of the locally fluctuating spontaneous magnetization of ferromagnetic materials.[82] The magnetization is coupled to stress by the magnetostriction constant, which is not zero for the nanocrystalline $Fe_{73.5}Si_{13.5}B_9Nb_3Cu_1$ alloy (it is ∼ 2 ppm only in the spatial average[51,83,84]). The heat treatment fails to completely relax stresses arising from the differences in the quenching rate between both sides of the ribbon during the planar flow casting process and from surface irregularities.[51,83] Thus, the complex domain structure present in the as-quenched amorphous state[80,83] transforms into a more ordered configuration with slight orientation variations. Microscopic observations of no-field annealed ribbons reveal patches of wide slab in-plane domains divided by 180° Bloch walls, together with very narrow regular stripe patterns of closure domains.[83,85,86]

Table II. Parameters of the anhysteretic magnetization curves parallel to the rolling direction of a grain-oriented silicon steel under different longitudinal compressive stresses: the optimized critical applied fields and corresponding reduced magnetizations for both magnetization components (H_{cri} and m_{cri}), the parameters of the double Langevin–Weiss function (M_{Si}, a_i, α_i), the material's saturation polarization (J_S), the dimensionless products used in Fig. 1 ($\alpha_i|M_{Si}|/(3a_i)$), and the products related to the pseudo-domains density ($\mu_0|M_{Si}|a_i = N_i k_B T$)

Longitudinal compressive stress		0 MPa		10 MPa		25 MPa		35 MPa		45 MPa		67 MPa	
Parameter	Units	Comp. 1	Comp. 2	Comp. 1	Comp. 2	Comp. 1	Comp. 2	Comp. 1	Comp. 2	Comp. 1	Comp. 2	Comp. 1	Comp. 2
H_{cri}	A/m	11	2264	134	2269	294	234	473	424	707	598	1119	903
m_{cri}		0.49	0.53	0.67	0.61	0.62	0.67	0.67	0.69	0.69	0.72	0.72	0.75
M_{Si}	kA/m	1,260	260	1,268	346	5,315	− 3,810	11,164	− 9,671	7,590	− 6,114	5,993	− 4,560
a_i	A/m	9.3	98.7	8.7	346.3	37.6	15.1	31.3	21.6	34.1	19.5	33.5	17.1
α_i		8.5E-06	− 8.0E-03	− 1.3E-04	− 6.7E-03	− 6.0E-05	7.4E-05	− 5.1E-05	5.4E-05	− 1.1E-04	1.2E-04	− 2.3E-04	2.4E-04
$J_S = \mu_0 \sum_i M_{Si}$	T	1.91	2.03	1.89	1.88	1.86	1.80						
$\alpha_i\|M_{Si}\|/(3a_i)$		0.38	–	− 6.14	− 2.22	− 2.81	6.21	− 6.02	7.98	− 8.46	12.60	− 13.77	21.74
$\mu_0\|M_{Si}\|a_i$	J/m^3	15	32	14	150	251	72	440	263	325	150	252	98

The grain-oriented electrical steel in the demagnetized state without applied stress presents wide domains parallel to the rolling direction.[40,54,71] When low compressive stresses are applied parallel to the rolling direction, the domain structure evolves into the so-called "stress pattern I", which exhibits narrower domains oriented almost transversely to the rolling direction.[54,71,87] As the stress increases, domains become smaller in both length and width:[87,88] at intermediate stress levels, Perevtov et al. observed a checkboard pattern, which ultimately evolved into a more complicated surface structure called "stress pattern II" (for compressive stresses over $\sim$ 25 MPa), which consists in a much-refined domain structure.[54,71] Although $N_i k_B T$ does not show a linear dependence on stress, distinct values for low, intermediate, and high levels of stress may be associated with the described patterns.

CONCLUSION

We have presented a physically-based theory to describe the ideal anhysteretic magnetization of soft magnets at a macroscale. The proposed equation of state, based on the mean-field theory, is simple and yet accurate over wide ranges of the applied field. We have addressed the most controversial issues of the model, including the definition of the anhysteretic curve itself, the possibility of having opposing component magnetizations in a nonhomogeneous material, and the possibility of having negative molecular fields in ferromagnets with uniform anisotropies.

The proposed set of plots helps to unravel the magnetic homogeneity of the material and to identify the number of component magnetizations to model. A semi-log magnetization derivative plot is central to this end and also helps the curve-fitting process.

We investigated examples of emerging and traditional soft magnetic materials for energy applications and other electromagnetic devices: a nanocrystalline alloy with different induced anisotropies and a grain-oriented electrical steel under increasing compressive stresses. These challenging modeling scenarios, successfully fitted for the first time, demonstrate the scientific and technical utility of our method. On the one hand, the model parameters were related to the physical properties of the materials and, although they were not calculated by an ab initio approach, they could be easily retrieved by the step-by-step method provided here. On the other hand, noisy data digitized from the literature could be accurately described by a single analytic function, with both $M(H)$ and $\partial M/\partial H(H)$ continuous and differentiable. Fits to more precise experimental data are clearly possible.

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CONFLICT OF INTEREST

On behalf of all authors, the corresponding author states that there is no conflict of interest.

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